

Software Engineering

Requirement Engineering

3 Ba INF
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Kasper Engelen
kasper.engelen@uantwerpen.be

1 Practical

- Deadline: **Sunday, October 4, 2026, 22u00**
- Hand in via **Blackboard** as `GroupX_requirementEngineering.zip`
- The customer requirements can be found on Blackboard (`customer_requirements.pdf`).

2 Context

In this session you will turn **customer requirements** into a more **detailed description** of the system. The requirements come in the form of a text. More specifically, you will learn how to

- read and **analyse customer requirements** to identify the needs of the customer,
- specify **actors and use cases**,
- specify **user stories**,
- **reflect** on the requirements document by coming up with additional questions,
- explore how **Generative AI** may support the above.

3 Report and evaluation

The evaluation will be **in-person**, where we will discuss your solutions together. You are free to balance the work across your team of 3 students. At the evaluation **every student is expected to understand and explain** the solution.

While making the assignments you should make use of the attached templates. When you have finished the assignments, you can hand them in via **Blackboard as a zip-file**. The templates are essential to ensure a **smooth evaluation** and to ensure that you do not **miss out on any grades!**

During this assignment you will explore how Generative AI can support software engineering activities. Two persons in the group will perform the assignments with traditional means (Section 4.1 — Section 4.4). The third person will use a Generative AI tool to work in parallel (Section 4.5). You will switch these roles over the course of the semester. Afterwards you will compare the results and reflect upon the use of Generative AI (Section 4.6).

Note that for the exams you will not have access to generative AI so you should be able to perform the tasks without such support.

4 Assignment

4.1 Use cases

In this exercise we will analyse the requirements and formalise them with use cases and actors. Use cases describe the **functionality** of the system, while actors describe who will **interact** with the system. In order to identify actors and use cases you can use the questions on Slide 16.

Assignment 1

Perform the following tasks:

1. carefully read the requirements document,
2. identify all **actors**,
3. identify 20 **use cases** (make sure all actors are covered),
4. give for every actor their **responsibilities**: what are they trying to achieve? (e.g. a customer wants to buy a car),
5. give for every actor their **goals** by linking them to use cases: how will they use the system to fulfill their responsibilities (e.g. a customer sorts all cars by price from high to low),
6. create an **table** with all use cases names and numbered them with UC-xx for easier reference.
7. create an **overview** of all actors and use cases like on Slide 17.

4.2 Scenarios

Now that we have a high-level overview of the functionalities of the system and the people interacting with the system, we can move towards a more **detailed specification** of the system. You will pick three use cases and create scenarios for them (both primary and secondary). Pay close attention to the following:

1. Make sure you specify the scenario using the scenario table template. The intended audience is the **system development team**. The granularity is **detailed** and on the level of this **specific functionality**.
2. Make sure the **goal** is specified: what exactly will this specific use-case do for the customer?
3. **Pre-condition**: what needs to be done/provided in order for this use-case to work?
4. **Success** end condition: if successful, what is the result of the use-case?
5. **Fail** end condition: under what conditions do we consider the use-case to have failed? What secondary scenario will you use to handle this?
6. All primary and secondary **actors** relevant to this use-case should be clearly identified.
7. What **exceptions** can occur during the use-case? How will this impact the actor? Can the actor do something to mitigate this exception, or will it lead to a fatal error?

Assignment 2

Carefully choose **three interesting use cases** that you will turn into scenarios. Do the following for each chosen use case:

1. Fill in the **scenario table** from the templates, keeping in mind the points of attention mentioned above.
2. Make sure you provide a detailed **step-by-step description** of the scenario.
3. When an **exception** occurs, you should provide a “branch” which deals with the **alternative execution path**. Indicate precisely which branches are related to an exception, and which belong to the normal operating of the system.

4.3 User stories

Next, we will formalise the requirements using a different technique: user stories. Read the requirements and identify possible users of the system as well as what they want to do with the system. Choose **three functionalities** and discuss the following questions to obtain more insight:

1. Identify the **user**: who is this user story aimed at?
2. Identify the **goal**: what does the user want to achieve by interacting with the system?

3. Identify the **benefit**: what will this user story contribute to? What is the user trying to achieve?
4. Provide **conditions of satisfaction**: what will the system need to do in order to complete the story?
5. How does this functionality compare to the **INVEST criteria**?

Note: for this exercise it might be useful to compare with the use cases of the previous exercises.

Assignment 3

Use the obtained insights to do the following exercises for each of the three chosen functionalities:

1. Carefully read the requirements document.
2. Create a **user story card** for each user story (examples on Slide 26).
3. Consider for each user story the **INVEST criteria** (Slide 27). Answer for each criterium why the user story conforms to it.

4.4 Customer interaction

In the previous exercises you might have noticed that the requirements provided by the client are not always clear or fully specified. As part of agile development, you will at some point in the future have the opportunity to talk to the client to ask for clarification.

In this last exercise you will reflect on the requirements document and the previous exercises to come up with interesting questions that could provide additional insight into what the client wants or what the system needs to look like.

Assignment 4

Come up with seven or more **interesting questions** that the developers can ask the customer. You can use the following guiding questions while discussing the requirements document (you may also come up with your own!):

- Is the current requirements document clear?
- Is it clear what the customer needs?
- Are all the requirements realistic? Is developing the system actually feasible?
- Do we have an idea of what the big picture is?
- Who will use the system?
- When, where, and in what setting will the system be used?

4.5 Same assignment using Generative AI

All previous assignments were performed by your team mates by hand. In this part, you will instead use Generative AI to perform a few of the tasks above.

We recommend that you make use of Microsoft Copilot, provided by the University, but you are also allowed to use another **paid** Generative AI tool such as Claude, ChatGPT, etc. Do not use AI tools that are free, since those are sub-optimal and do not reflect the tools you will use in your later job.

Assignment 5

Complete the following tasks:

- Ask the Generative AI to create the use-cases (Section 4.1) or the scenarios (Section 4.2) exercise.
- Try to fool the generative AI by requesting scenarios for use cases which don't exist.
- Ask the AI to derive user stories (Section 4.3).
- Ask the AI to perform the customer interactions exercise (Section 4.4). Which non-obvious questions would you now ask to the clients to refine the requirements?
- Some Generative AI tools will suggest a few follow-up tasks after each answer. Choose one of these.

For the report, first tell us who performed the manual part and who performed supported with Generative AI; you need to switch between these roles over the course of the semester. Then tell us which Generative AI tool you used and how long your chat session lasted. If possible also report how many tokens you used.

Then for each task, report the following:

- What prompt techniques (zero-shot, few-shot, chain-of-thought, ...) did you use? Provide the exact prompt(s) as well.
- Did you have to refine the prompt? If yes explain why this was needed and how you improved?
- How did you verify whether the response was good, acceptable, correct, useful ...?

4.6 Compare and Reflect

Compare the results from the manual assignments against the ones done with Generative AI support and reflect.

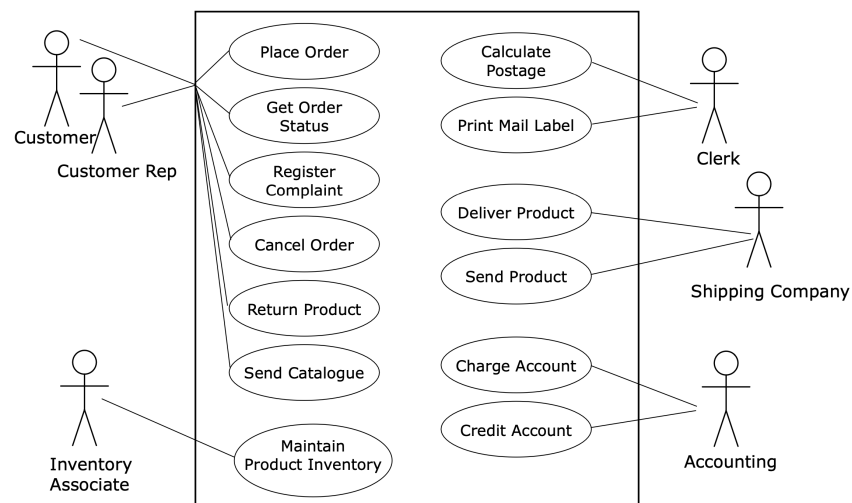
Assignment 6

- Where did the list of use-cases (Section 4.1) differ?
- Which of the scenarios (Section 4.2) were easier to read? Which ones were incomplete?
- Which of the user stories (Section 4.3) were easier to read? Were there conditions of satisfaction which were missing?
- Did the Generative AI help to find relevant questions for the customer interaction (Section 4.4)?
- Did the Generative AI hallucinate?
- Where and how would you use Generative AI during requirements engineering? How does Generative AI complement the human-in-the-loop? Where would you **not** use it.

5 Templates

Below you will find a number of templates that you should use for the assignment. You can re-create these templates using a program of choice (LaTeX, MS Word, ...).

5.1 Use case and actors diagram (Slide 17)



5.2 Scenario table (Slides 22 and 23)

UCxx – Use case name	
Audience	System development team
Level	specific functionality
Granularity	detailed
Preconditions	...
Success condition	...
Fail condition	...
Primary actors	...
Secondary actors	...
Trigger	...

Step	Action
1.	Description of step 1
2.	Description of step 2
3.	Description of step 3
Branch	Subvariations
2.	Different options on step 2 ...
3.	Different options on step 3 ...
Branch	Exception
2.	Exception on step 2 ...
any	Exception that can occur at any step ...

5.3 Story card (Slide 26)

